

First-Order ODEs (Zill 2.2–2.5)

Separable: $\frac{dy}{dx} = g(x)h(y)$, so $\int \frac{dy}{h(y)} = \int g(x) dx + C$ – separate variables, then integrate both sides.

Linear & integrating factor: $\frac{dy}{dx} + P(x)y = Q(x)$, $\mu(x) = e^{\int P(x) dx}$, $y = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x) dx + C \right]$ – multiplying by μ makes the LHS $\frac{d}{dx}[\mu y]$.

Exact equations: $M(x, y) dx + N(x, y) dy = 0$ is exact iff $M_y = N_x$; solution $f(x, y) = \int M dx + g(y) = C$ with $f_y = N$ – potential function f has $f_x = M$, $f_y = N$.

Integrating factors for non-exact equations: $\mu(x) = e^{\int (M_y - N_x)/N dx}$ if that ratio depends on x only; $\mu(y) = e^{\int (N_x - M_y)/M dy}$ if it depends on y only – multiplying by μ restores exactness; try the x -only case first.

Homogeneous-coefficient DE (sub $y = vx$): $\frac{dy}{dx} = f(y/x)$, $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$ – used when M, N are homogeneous of the same degree; reduces to a separable equation in v, x .

Bernoulli equation: $\frac{dy}{dx} + P(x)y = Q(x)y^n$, $n \neq 0, 1$, sub $w = y^{1-n} \Rightarrow \frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$ – turns it into a first-order linear equation in w .

Linear Higher-Order ODEs (Zill 3.1–3.6)

Reduction of order: given one solution y_1 of $y'' + Py' + Qy = 0$, $y_2 = y_1(x) \int \frac{e^{-\int P dx}}{[y_1(x)]^2} dx$ – produces a second, linearly independent solution $y_2 = y_1 v(x)$.

Homogeneous, constant coefficients: $ay'' + by' + cy = 0 \Rightarrow am^2 + bm + c = 0$ (sub $y = e^{mx}$ gives the auxiliary equation). Three root cases:

Distinct real m_1, m_2 : $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$.

Repeated $m_1 = m_2$: $y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$.

Complex $m = \alpha \pm \beta i$: $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$.

n th-order, repeated roots: a real root m repeated k times contributes $e^{mx}, x e^{mx}, \dots, x^{k-1} e^{mx}$; a complex pair $\alpha \pm \beta i$ repeated k times contributes $e^{\alpha x} x^j \cos \beta x$ and $e^{\alpha x} x^j \sin \beta x$ for $j = 0, \dots, k-1$.

Undetermined coefficients – trial y_p forms: $g(x) = k$ (const) $\rightarrow A$; degree- n polynomial $\rightarrow A_n x^n + \dots + A_1 x + A_0$; $k e^{ax} \rightarrow A e^{ax}$; $k \cos \beta x$ or $k \sin \beta x \rightarrow A \cos \beta x + B \sin \beta x$ (always include both); product $\rightarrow$ product of matching forms (e.g. $x e^{ax} \cos \beta x$). If any assumed term duplicates a term in y_c , multiply the whole y_p guess by x (or x^2) until the duplication is gone.

Variation of parameters (2nd order): write $y'' + Py' + Qy = f(x)$ in standard form, $y_c = c_1 y_1 + c_2 y_2$, Wronskian $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$, $u_1' = \frac{-y_2 f(x)}{W}$, $u_2' = \frac{y_1 f(x)}{W}$, $y_p = u_1 y_1 + u_2 y_2$.

Cauchy-Euler equation (equidimensional), $x > 0$: $ax^2 y'' + bxy' + cy = 0$, sub $y = x^m \Rightarrow am(m-1) + bm + c = 0$. Distinct real: $y = c_1 x^{m_1} + c_2 x^{m_2}$; repeated m_1 : $y = c_1 x^{m_1} + c_2 x^{m_1} \ln x$; complex $\alpha \pm \beta i$: $y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$.

General solution of a non-homogeneous linear DE: $y = y_c + y_p$ – y_c (complementary function) solves the associated homogeneous equation; y_p (particular solution) is any one solution of the full equation.

Laplace Transforms (Zill 4.1–4.5)

Definition: $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$, $t \geq 0$ – converges for s large enough provided f is piecewise continuous and of exponential order. Linearity: $\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$.

Laplace transform pairs:

$$\mathcal{L}\{1\} = 1/s, s > 0, \quad \mathcal{L}\{t\} = 1/s^2, s > 0$$

$$\mathcal{L}\{t^n\} = n!/s^{n+1}, n = 1, 2, 3, \dots, s > 0$$

$$\mathcal{L}\{e^{at}\} = 1/(s-a), s > a$$

$$\mathcal{L}\{\sin kt\} = k/(s^2 + k^2), \quad \mathcal{L}\{\cos kt\} = s/(s^2 + k^2), s > 0$$

$$\mathcal{L}\{\sinh kt\} = k/(s^2 - k^2), \quad \mathcal{L}\{\cosh kt\} = s/(s^2 - k^2), s > |k|$$

$$\mathcal{L}\{t \sin kt\} = 2ks/(s^2 + k^2)^2, s > 0$$

$$\mathcal{L}\{t \cos kt\} = (s^2 - k^2)/(s^2 + k^2)^2, s > 0$$

$$\mathcal{L}\{e^{at} \sin kt\} = k/[(s-a)^2 + k^2], s > a$$

$$\mathcal{L}\{e^{at} \cos kt\} = (s-a)/[(s-a)^2 + k^2], s > a$$

$$\mathcal{L}\{e^{at} t^n\} = n!/(s-a)^{n+1}, s > a$$

$$\mathcal{L}\{\mathcal{U}(t-a)\} = e^{-as}/s, s > 0, a \geq 0$$
 – unit step

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}, a \geq 0$$
 – Dirac delta

Inverse Laplace transform: $\mathcal{L}^{-1}\{F(s)\} = f(t)$, linear: $\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F(s)\} + b\mathcal{L}^{-1}\{G(s)\}$ – use partial fraction decomposition to match table entries, then invert term by term.

Transform of derivatives: $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$; $\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$; $\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - \dots - f^{(n-1)}(0)$ – turns linear DEs with initial conditions into algebraic equations in s .

Transform of an integral: $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$ – dividing by s corresponds to integrating f from 0 to t .

First translation theorem: $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$ – multiplying $f(t)$ by e^{at} shifts the transform: replace every s in $F(s)$ by $(s-a)$.

Second translation theorem: $\mathcal{U}(t-a) = 0$ for $t < a$, 1 for $t \geq a$. $\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as} F(s)$; if $g(t)$ is not already a shift of a known function, use $\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as} \mathcal{L}\{g(t+a)\}$.

Transform of a periodic function: $\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$ – valid when $f(t+T) = f(t)$ for all $t \geq 0$; only one period needs to be integrated.

Dirac delta (unit impulse): $\int_0^\infty \delta(t-a) dt = 1$, $\int_0^\infty f(t) \delta(t-a) dt = f(a)$; $\mathcal{L}\{\delta(t-a)\} = e^{-as}$, $a \geq 0$; at $a = 0$, $\mathcal{L}\{\delta(t)\} = 1$.

Convolution theorem: $(f*g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau$, and $\mathcal{L}\{f*g\} = F(s)G(s)$ – convolution in t is ordinary multiplication in s ; useful for inverting a product of two known transforms.

Quick Reference – First-Order ODE Method by Type

Separable ($dy/dx = g(x)h(y)$) – separate variables, integrate both sides.

Linear ($y' + P(x)y = Q(x)$) – integrating factor $\mu = e^{\int P dx}$.

Exact ($M dx + N dy = 0$, $M_y = N_x$) – find potential f with $f_x = M$, $f_y = N$; solution $f(x, y) = C$.

Non-exact – multiply by an integrating factor $\mu(x)$ or $\mu(y)$ to restore $M_y = N_x$, then solve as exact.

Homogeneous ($dy/dx = f(y/x)$) – sub $y = vx$, reduces to separable equation in v .

Bernoulli ($y' + Py = Qy^n$) – sub $w = y^{1-n}$, reduces to linear equation in w .

Quick Reference – Root-Case Forms Side by Side

Constant-coefficient $ay'' + by' + cy = 0$ (auxiliary eqn. $am^2 + bm + c = 0$) vs. Cauchy-Euler $ax^2y'' + bxy' + cy = 0$ (auxiliary eqn. $am(m-1) + bm + c = 0$, sub $y = x^m$):

Distinct real m_1, m_2 – const. coeff.: $y = c_1e^{m_1x} + c_2e^{m_2x}$;
 Cauchy-Euler: $y = c_1x^{m_1} + c_2x^{m_2}$.

Repeated m_1 – const. coeff.: $y = c_1e^{m_1x} + c_2xe^{m_1x}$; Cauchy-Euler: $y = c_1x^{m_1} + c_2x^{m_1} \ln x$.

Complex $\alpha \pm \beta i$ – const. coeff.: $y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$;
 Cauchy-Euler: $y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$.

Quick Reference – Solving an IVP by Laplace Transform

1) Apply $\mathcal{L}\{\cdot\}$ to both sides; use $\mathcal{L}\{y^{(n)}\} = s^n Y(s) - s^{n-1}y(0) - \dots - y^{(n-1)}(0)$ to convert the DE into an algebraic equation in $Y(s) = \mathcal{L}\{y\}$.

2) Solve algebraically for $Y(s)$.

3) Decompose $Y(s)$ by partial fractions into pieces matching table entries; apply the shifting theorems for terms with e^{as} or e^{-as} factors, or the convolution theorem for a leftover product $F(s)G(s)$.

4) Invert term by term: $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.